LLM-Guided Drone Path Planning

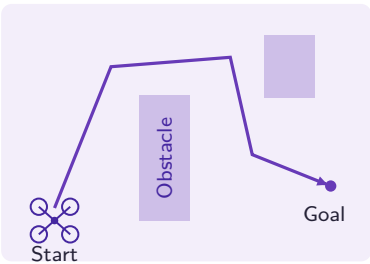
A critical discussion of Farooq et al.

Arush Ram Anandh

Paper: *Goal-Oriented Intelligence Path Planning for LLMs Based Drone in Adversarial Conditions*

Can foundation models help drones navigate?

- Reach a destination while avoiding obstacles.
- Unfamiliar or changing environments motivate adaptive navigation.
- The paper explores LLMs/VLMs for waypoint planning using visual context.



Illustrative route, not an experimental result

Research question

What useful role can a foundation model play in a drone navigation system?

Source: `paper.pdf`, pp. 1, 3–4.

Why LLMs? The authors' broader motivation



Coordination

Multi-UAV communication, mission planning and information exchange.



Contextual reasoning

Interpret instructions, observations, unexpected events and anomalies.



Adaptation

Respond to changing obstacles with goal-directed navigation decisions.

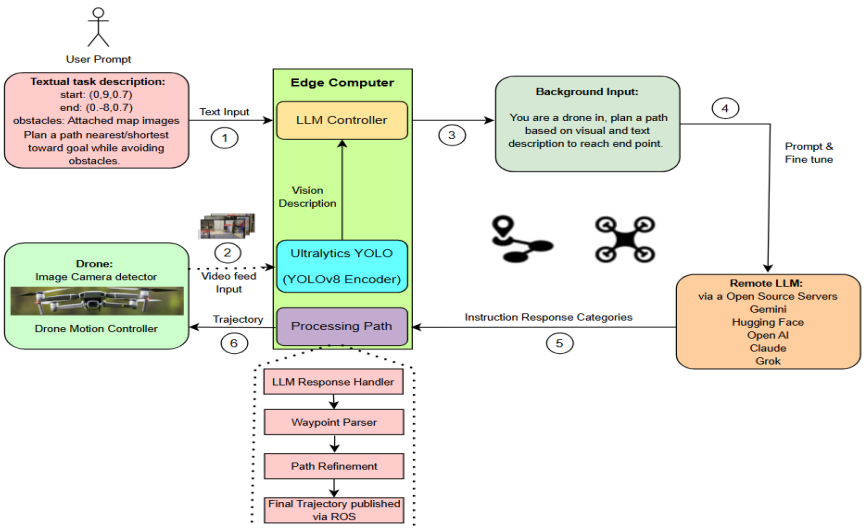
These capabilities motivate the paper. Its experiments cover a narrower task.

Specific experiment

Vision-guided waypoint generation in simulated obstacle environments.

Source: `paper.pdf`, p. 3, broader motivation; pp. 4–5, experimental scope.

Proposed architecture: observations to flight



Original Figure 3, cropped from the paper

Source: `paper.pdf`, p. 4, Fig. 3 and methodology; p. 5, RGB/depth observations.

Reading guide

1. Start, goal, instructions and camera input
2. YOLOv8 object detection
3. Constructed input to remote LLM/VLM
4. Parse and interpolate waypoints
5. ROS to motion controller

Not specified: the exact arrangement of the fine-tuned and remotely prompted models.

Fine-tuning, prompt structure and inference

Training

Simulated start, goal and obstacles

↓

RRT-generated reference waypoints

↓

Fine-tune a pretrained LLM
Cross-entropy loss

Inference

Start, goal, visual context, instruction

↓

LLM/VLM generates waypoint sequence

↓

Parse, interpolate and control

$$\underbrace{\text{Input} = \text{Start} + \text{Goal} + \text{Visual Map} + \text{Instruction}}_{\text{Paper's input expression, abbreviated from Eq. (1)}}$$

Reported: 105 epochs; training loss ≈ 0.15 near epoch 65; three visual-map examples for few-shot prompting.

Not specified: model identity/method, dataset size/split, coordinate formatting, exact prompt, visual-to-spatial conversion, or which model produced the aggregate results.

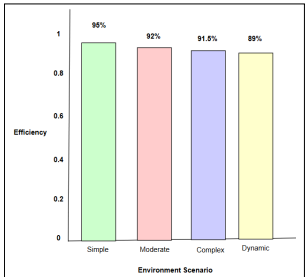
Inconsistency: reference-path clearance is 2 m on p. 4 and 1.2 m on p. 5.

Source: `paper.pdf`, pp. 4–5, methodology, Eq. (1) and training description. Flows summarize the prose.

Experimental results: what was measured?

Table I: reported simulation results

Environment	Obstacles	Moving obstacles	Success rate	Mean time (s)
Simple	8–10	3	95.0%	65.7
Moderate	10–12	7	92.0%	65.4
Complex	12–16	10	91.5%	69.2
Dynamic	16–20	12	89.0%	72.0



Original Figure 4

Success decreases from **95% to 89%** across the reported categories.

Evidence limits

All four categories list moving obstacles. Formal difficulty definitions and obstacle motion models are unspecified.

Trial counts and uncertainty are unreported. Figure 4 calls success rates "efficiency" without defining a separate path-efficiency metric.

Source: `paper.pdf`, p. 5, Table I and Fig. 4; p. 6, conclusion claims.

What is missing from the evaluation?

Dimension	Missing or unclear
Reproducibility	Model identity, dataset size, trial count and splits
Baselines	Controlled comparisons with RRT/RRT*, A* or other planners
Safety	Collision rates, segment validation and measured clearance
Path quality	Length, optimality, smoothness and energy
Real-time operation	Inference latency, replanning frequency and compute
Generalization	Unseen environments, sensor noise and physical flight

What advantage does the learned planner demonstrate over the planner used to generate its reference paths?

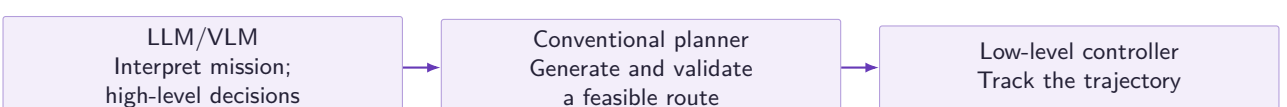
Source: `paper.pdf`, pp. 4–6. Evaluation gaps identified in this discussion.

Alternative planning methods and extensions

Our analysis / proposed extensions

Approach	Role	Why include it?
A*, RRT, RRT*	Classical geometric planning	Feasibility, path cost and a reproducible baseline
Model Predictive Control (MPC)	Receding-horizon planning/control	Dynamic obstacles and control constraints
Reinforcement Learning (RL)	Learned navigation policies	Learning from interaction and adaptation
Hybrid LLM + classical planner	Semantic decisions with geometric validation	Test added value while retaining explicit safety checks

Proposed hybrid pipeline



Source: Our analysis, motivated by the omissions in `paper.pdf`, pp. 4–6; this pipeline is not evaluated in the paper.

Conclusion: promise versus evidence

What the paper offers

An integration concept linking visual perception, LLM/VLM planning, waypoint processing and drone control.

Preliminary simulated goal-reaching rates across four obstacle categories.

What it does not establish

Reproducible fine-tuning and inference details.

Comparative benefit over established planners.

Safety, real-time feasibility and real-world generalization.

Foundation models may add value at the high-level decision layer, but that value needs controlled evaluation against conventional planning methods.

Source: `paper.pdf`, pp. 4–6; final assessment is our analysis.